

Producer Consumer Problem - C Program

C Program

```
#include <stdio.h>

int mutex = 1;
int full = 0;
int empty = 3;
int x = 0;

int main() {
    int n, ch;

    printf("1. Producer\n2. Consumer\n3. Exit\n");

    while(1) {
        printf("Enter your choice: ");
        scanf("%d", &ch);

        switch(ch) {
            case 1:
                if((mutex == 1) && (empty != 0)) {
                    mutex = 0;
                    full++;
                    empty--;
                    x++;
                    printf("Producer produces item %d\n", x);
                    mutex = 1;
                }
                else {
                    printf("Buffer is Full\n");
                }
                break;

            case 2:
                if((mutex == 1) && (full != 0)) {
                    mutex = 0;
                    printf("Consumer consumes item %d\n", x);
                    full--;
                    empty++;
                    x--;
                    mutex = 1;
                }
                else {
                    printf("Buffer is Empty\n");
                }
                break;

            case 3:
                return 0;
        }
    }

    return 0;
}
```

Sample Input / Output

Sample Input:

```
1
1
2
3
```

Sample Output:

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Producer produces item 1

Enter your choice: 1
Producer produces item 2

Enter your choice: 2
Consumer consumes item 2
Enter your choice: 3
```